ECE 2305 – Introduction to C Programming

Programming Tutorial 2

Write a C++ Program to investigate the Basic Data Types in C++. Declare a variable of type **int**, **double**, **char** and **bool** and assign a value to each of the variables. Determine the address of each variable using the **&** operator and the memory requirement using the **sizeof** keyword. Use the following bit of code as an example for type **int** data.

Main Code

```
//ECE 2305 Tutorial 02 - Basic Data Types in C++
#include <iostream>
using namespace std;

int main()
{
	cout << "Basic Data Types C++" << endl;
	cout << endl;

	int IntVar = 100;
	cout << "The address of IntVar is: " << &IntVar << endl;
	cout << "The IntVar requires " << sizeof(IntVar) << " bytes" <<
endl;
	cout << "The value of IntVar is: " << IntVar << endl;
	cout << endl;

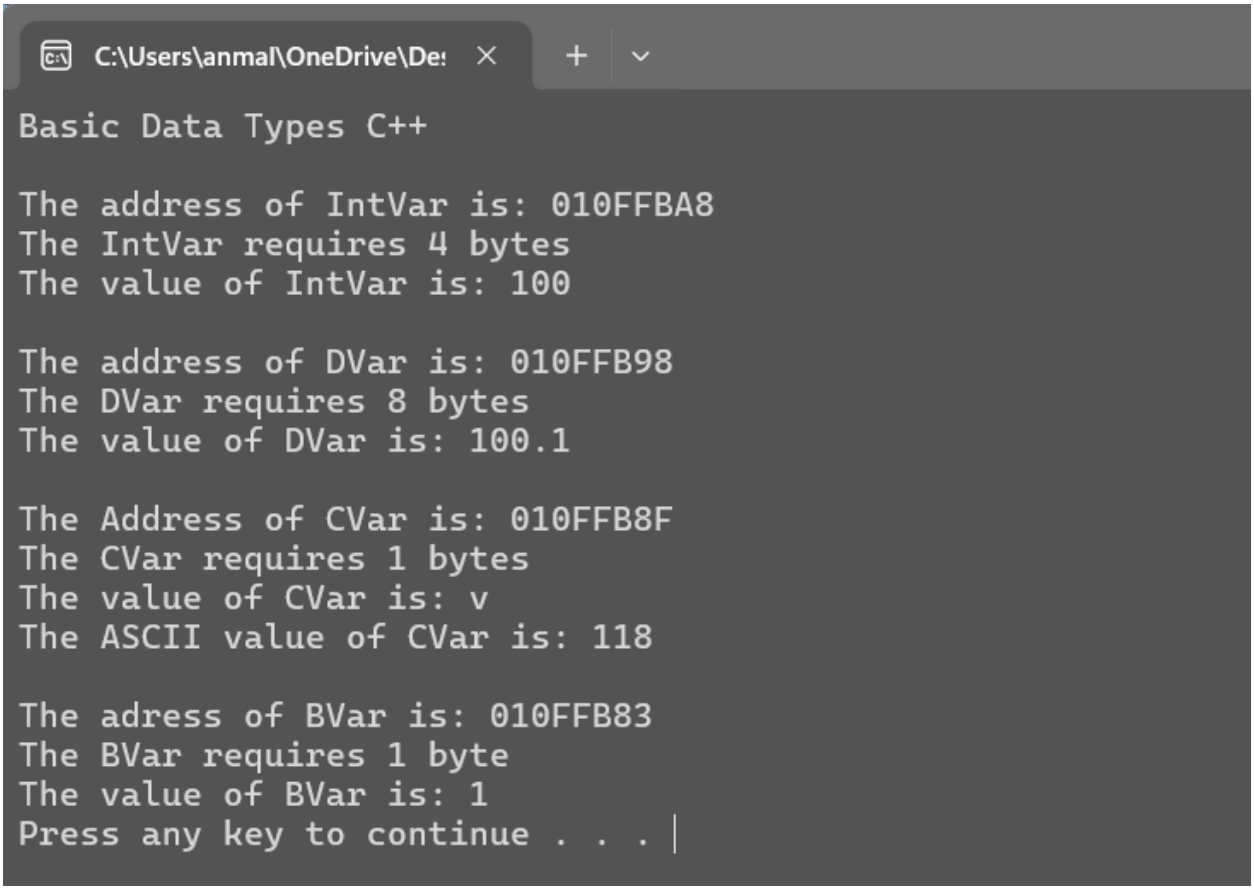
	double DVar = 100.1;
	cout << "The address of DVar is: " << &DVar << endl;
	cout << "The DVar requires " << sizeof(DVar) << " bytes" << endl;
	cout << "The value of DVar is: " << DVar << endl;
	cout << endl;

	char CVar = 'v';
	cout << "The Address of CVar is: " << (void*)&CVar << endl;
	cout << "The CVar requires " << sizeof(CVar) << " bytes" << endl;
	cout << "The value of CVar is: " << CVar << endl;
	cout << "The ASCII value of CVar is: " << int(CVar) << endl;
	cout << endl;

	bool BVar = 1;
	cout << "The adress of BVar is: " << &BVar << endl;
	cout << "The BVar requires " << sizeof(BVar) << " byte" << endl;
	cout << "The value of BVar is: " << BVar << endl;
	cout << endl;

	system("pause");
	return 0;
}
```

Output


C:\Users\anmal\OneDrive\De:
Basic Data Types C++
The address of IntVar is: 010FFBA8
The IntVar requires 4 bytes
The value of IntVar is: 100
The address of DVar is: 010FFB98
The DVar requires 8 bytes
The value of DVar is: 100.1
The Address of CVar is: 010FFB8F
The CVar requires 1 bytes
The value of CVar is: v
The ASCII value of CVar is: 118
The adress of BVar is: 010FFB83
The BVar requires 1 byte
The value of BVar is: 1
Press any key to continue . . .